

MUHAMMAD HISYAM

Full Stack & AI Software Engineer

Email: muhammad2504hisyam@gmail.com · Location: Bekasi, Indonesia

GitHub: github.com/caleheinz25 · LinkedIn: linkedin.com/in/muhammad-hisyam-65502332a

Portfolio: mhisyam.com



ABOUT ME

Fresh Graduate Computer Science Engineer from **Universitas Gunadarma** specializing in cross-platform desktop architecture, reactive web applications, medical deep learning pipelines, and agentic AI integrations. Creator of 26+ open-source and enterprise-grade software projects built with **.NET / C#, Rust (Tauri), TypeScript, Python (PyTorch), Golang, Java, and Supabase/PostgreSQL**.

SKILLS & TECH STACK

Languages: C# TypeScript JavaScript Rust Python Golang Java PHP SQL

Frontend & Desktop: Tauri Astro 5 React.js SolidJS SvelteKit Next.js Tailwind CSS 4 TanStack Router

Backend & Cloud: .NET Supabase Node.js Express.js Go (Gin/Fiber) Spring Boot Laravel REST APIs WebSockets

AI & Machine Learning: PyTorch U-Net OpenCV Model Context Protocol (MCP) Gemini API

Databases & DevOps: PostgreSQL Supabase RLS MongoDB Redis Docker Git Vite Bun Linux CLI

FEATURED PROJECTS

RetailHub – Retail POS & Inventory Management Platform

[GitHub Repo]

Tech Stack: Tauri Rust TypeScript Supabase TanStack Router Tailwind CSS

- Engineered a high-performance cross-platform desktop POS & inventory management system using Tauri (Rust backend) and Vite.
- Implemented real-time database synchronization via Supabase PostgreSQL, Row Level Security (RLS) policies, and custom PL/pgSQL database functions.

Developer Portfolio & Interactive Dev Suite

[GitHub Repo]

Tech Stack: Astro 5 SolidJS TypeScript Tailwind CSS 4 Unix Shell CLI

- Built an interactive developer portfolio featuring an embedded retro Unix terminal shell (`hisyam-cli`), REST API response client, audio player, and 8 mini-games arcade.
- Optimized performance achieving 100% lighthouse scores via Astro static page generation and SolidJS component hydration.

Pneumonia Medical Image Segmentation U-Net

[GitHub Repo]

Tech Stack: Python PyTorch U-Net OpenCV Deep Learning

- Designed a deep learning semantic segmentation pipeline in PyTorch for detecting infected lung regions in chest X-ray medical scans.
- Automated dataset preprocessing, data augmentation, loss tracking, and IoU / Dice metric evaluations.

Gemini Model Context Protocol (MCP) Client

[\[GitHub Repo\]](#)

Tech Stack:

TypeScript

Node.js

MCP SDK

Gemini API

Agentic AI

- Developed a modular MCP client enabling agentic AI workflows, tool calls, filesystem access, and server integrations with Gemini.

Golang & SvelteKit Real-Time Chat App

[\[GitHub Repo\]](#)

Tech Stack:

Go

WebSockets

SvelteKit

Tailwind CSS

- Implemented a low-latency concurrent chat application powered by a Golang WebSocket server backend and a reactive SvelteKit interface.

Secure Java JWT Authentication Service

[\[GitHub Repo\]](#)

Tech Stack:

Java

Spring Boot

JWT

PostgreSQL

- Built an enterprise authentication microservice featuring access/refresh tokens, sliding expiration, and instant version revocation.

EDUCATION & HIGHLIGHTS

Universitas Gunadarma

Bekasi, Indonesia

Fresh Graduate

Bachelor of Computer Science (Teknik Informatika) — Fresh Graduate

- Maintainer of 26+ open-source GitHub repositories spanning AI, Web, Desktop, and Microservices.
- Active contributor to community and open-source software solutions including SmartAssist AI and Woven.